

FINAL PORTFOLIO

MATH 141 : LINEAR ANALYSIS

Sharlize Berrospe

1 EVALUATE : TRUE OR FALSE

2

- 1.) Two subspaces V and W can be orthogonal without being orthogonal complements

Ch. 3

Their dimensions can be too small. Only really fits when dimensions of both are right. In solving this question, the dimensions of a subspace may easily be overlooked and not taken into account, which is essential.

- 2.) Consider every matrix leading to a linear transformation. Then, every linear transformation leads to a matrix.

Ch. 2

By the properties of linearity: $A(cx + dy) = cA(x) + dA(y)$, every transformation that satisfies this is a linear transformation. An $A_{m \times n}$ matrix transforms vectors in $\mathbb{R}^n \rightarrow \mathbb{R}^m$. Since the original vector x has n components, and the transformed vector Ax has m components. Any vector space allows the combination $cx + dy$, wherein the "vectors" are $\vec{x}$ and $\vec{y}$, but could also be polynomials, matrices, or functions $x(u)$, $y(u)$. It is much easier to prove and make sense of matrices leading to L.T. than the latter half of the statement. Therefore, it might be assumed to be a trick question, and there are hidden "rules" at play.

- 3.) Complex numbers are not needed for real matrices.

Ch. 5

If there are not enough real eigenvalues in a matrix, there will always be n complex eigenvalues. Similarly, if there are too few eigenvectors in $\mathbb{R}^n$, we look for some in $\mathbb{C}^n$, the complex plane. It is not likely for someone to correctly guess complex values hold importance in real cases, or that there is even such a thing as too few [real] eigenvalues or eigenvectors.

- 4.) For a matrix A , if its determinant, $\det(A)$, is equal to zero, A is invertible. Similarly, if $\det(A) \neq 0$, A is singular.

Ch. 4

The opposite is true; we use $\det(A - \lambda I) = 0$ to find for which eigenvalues is $A - \lambda I$ singular. When $\det(A) = 0$, the columns of A are not linearly independent, and $\det(A)$ = the volume of a box in n -dimensional space. From the statement, it is easy to conclude that is true, as often times in math, something resulting in zero is the go-ahead for specific assumptions we are looking to be true.

- 5.) A matrix A is invertible if $Ax = 0$ and $x \neq 0$.

Ch. 1

No matrix A^{-1} can multiply the zero vector, Ax , and still end up with a nonzero vector $\vec{x}$. A^{-1} would have to get back from $Ax = 0$ to x - impossible. It can be easy for someone to think it is possible for $Ax = 0$ with $x \neq 0$, so long as $A = 0$, but this cannot be the case. This is dependent on A and its linear independence, not so much focus on x .

- 3 These problems each tackle the various concepts from ch. 1-5 of the content covered in this course. The core understanding of invertibility and matrix properties, linear transformations, orthogonality and fundamental subspaces, determinants, and eigenvalues/vectors in complex/real components is essential to linear analysis (which these questions cover). These topics open the door to many different routes math can be taken and explored. Thus mastering the basics and properties now is pertinent to progressing in the field, no matter which route one may take.

1

ANSWER KEY:

1) TRUE	Ch. 3
2) TRUE	Ch. 2
3) FALSE	Ch. 5
4) FALSE	Ch. 4
5) FALSE	Ch. 1

1

a) From Discussion Wkst 4, #2: Find the complete solution $\vec{x} = \vec{x}_p + \vec{x}_n$.

$$x + 3y + 3z = 1$$

b) For this problem, we must find the particular soln. $\vec{x}_p$, and the soln. $\vec{x}_n$ by forming the augmented matrix from the system of equations given, row reducing, and finding the free variables.

$$2x + 6y + 9z = 5$$

$$-x - 3y + 3z = 5$$

2 At the time, although I fully understood how to form the augmented matrix and row reduce, I didn't really understand what a free variable was, and how to properly apply it to our soln. I confused the free variables with the pivot variables when expressing the latter in terms of the former, resulting in an incorrect complete soln. I also did not understand what the complete soln. meant graphically.

3

To figure out how to correctly express pivot variables in terms of the free variables, I learned it confused me to bring in additional variables in place of our free variables (e.g., in class we often use s and t). Once I started doing this, I stopped mixing up certain variables. I also believed it should be the free variables expressed in terms of pivot variables, as free are independent. To fix this, I did any additional problems on the topic I could find and would explain in writing and aloud each step of the way why I did what I did, to ingrain this. I also then learned in discussion what our complete soln. meant, graphically, and was able to remember that quickly.

4

$$\begin{cases} x + 3y + 3z = 1 \\ 2x + 6y + 9z = 5 \\ -x - 3y + 3z = 5 \end{cases} \Rightarrow \begin{bmatrix} 1 & 3 & 3 \\ 2 & 6 & 9 \\ -1 & -3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 5 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 1 & 3 & 3 & 1 \\ 2 & 6 & 9 & 5 \\ -1 & -3 & 3 & 5 \end{array} \right] \xrightarrow{R_2 - 2R_1, R_3 + R_1} \left[\begin{array}{ccc|c} 1 & 3 & 3 & 1 \\ 0 & 0 & 3 & 3 \\ -1 & -3 & 3 & 5 \end{array} \right] \xrightarrow{R_3 + R_2} \left[\begin{array}{ccc|c} 1 & 3 & 3 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 6 & 6 \end{array} \right] \xrightarrow{R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 3 & 3 & 1 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

pivot: x, z
free: y

$$\begin{aligned} 3z &= 3 \\ z &= 1 \end{aligned}$$

Let's say $\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

set $y=0$

$$x_n = y \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}, \quad x_p = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{aligned} x + 3y + 3z &= 1 \\ x + 3y + 3 &= 1 \\ x &= -3y - 2 \end{aligned}$$

$$\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3y - 2 \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$$

$$\vec{x} = \vec{x}_p + \vec{x}_n = \begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix} + y \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}; y \in \mathbb{R}$$

The line in $\mathbb{R}^3$ is described by $\begin{bmatrix} -3 \\ -3 \\ 1 \\ 0 \end{bmatrix}$ such that it passes through $\begin{bmatrix} -2 \\ 0 \\ 1 \end{bmatrix}$.

1

- a) From Discussion Wkst 2, #4. Let $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix}$.
- (a) Find the least-squares soln. $\hat{x}$ to $Ax = \vec{b}$.
- (b) Find the projection of $\vec{b}$ onto $C(A)$.
- (c) Compute $\|A\hat{x} - b\|^2$.
- b) For this problem we use what we know about the least-squares soln., $A^T A x = A^T b$, and the projection of b onto $C(A)$, $Ax = A(A^T A)^{-1} A^T b$.
- (d) Find a vector $\hat{q} \in \mathbb{R}^2$ different from $\hat{x}$.
Compute $\|A\hat{q} - b\|^2$ and compare your answer to that in part (c).

2

The part of this problem I struggled with was part d, as I didn't understand what vector $\hat{q} \in \mathbb{R}^2$ to choose or how that changed anything. I also did not understand what we were solving for with $\|A\hat{x} - b\|^2$, conceptually.

3

In familiarizing myself with the topic more in depth with the textbook, and in discussion, I learned that the $\hat{x}$ in $\|A\hat{x} - b\|^2$ works to minimize the squared error $\|A\hat{x} - b\|^2 = E^2$, and the vector $q = A\hat{x}$ is as close as possible to b , which means the vector that best fits the data. Graphically, these errors are the vertical distances to the straight line, and are squared, summed, and minimized.

4

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \text{ and } b = \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix}$$

(a) Least-Squares soln.: $A^T A x = A^T b$

$$\begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix}$$

$$\begin{bmatrix} 4+1+1 & 2+2+1 \\ 2+2+1 & 1+4+1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4+0-3 \\ 2+0-3 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 5 \\ 5 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 5 & | & 1 \\ 5 & 6 & | & -1 \end{bmatrix} R_2 - R_1$$

$$\begin{bmatrix} 1 & -1 & | & 2 \\ 5 & 6 & | & -1 \end{bmatrix} R_2 - 5R_1$$

$$\begin{bmatrix} 1 & -1 & | & 2 \\ 0 & 11 & | & -11 \end{bmatrix} \quad \begin{array}{ll} x_1 - x_2 = 2 & \|x_2 = -1 \\ x_1 - (-1) = 2 & x_2 = -1 \end{array}$$

$$x_1 = 1 \rightarrow \hat{x} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$(b) Ax = A(A^T A)^{-1} A^T b$$

$$= \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \left(\begin{bmatrix} 6 & 5 \\ 5 & 6 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 6/11 & -5/11 \\ -5/11 & 6/11 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{6}{11} + \frac{5}{11} \\ -\frac{5}{11} - \frac{6}{11} \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \} = Ax \checkmark$$

$$\text{proj}_{C(A)} b = Ax = \begin{bmatrix} 2 & -1 \\ 1 & -2 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$(c) \|A\hat{x} - b\|^2 = \sqrt{(-1)^2 + (-1)^2 + (-3)^2}^2$$

$$= 1 + 1 + 9 = 11$$

$$Ax - b = \begin{bmatrix} 1-2 \\ -1 \\ -3 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ -3 \end{bmatrix}$$

(d) let's say $\hat{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ any $\hat{y}$, as long as $\hat{y} \neq \hat{x} \rightarrow$ to prove $\|A\hat{x} - b\|^2$ minimizes as much as possible

$$A\hat{y} - b = \begin{bmatrix} 2 & 1 \\ 1 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix} = \begin{bmatrix} 2+1 \\ 1+2 \\ 1+1 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 0 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 5 \end{bmatrix}$$

$$\|A\hat{y} - b\|^2 = \sqrt{(1)^2 + (3)^2 + (5)^2}^2 = 1 + 9 + 25 = 35$$

As expected, $\|A\hat{x} - b\|^2 < \|A\hat{y} - b\|^2$,
and $\|A\hat{x} - b\|^2$ will be smaller than any squared
error we get with any vector $\hat{y}$.

1
a) From the textbook 4.3, #31:

Compute the determinants S_1, S_2, S_3 of these $1, 3, 1$ tridiagonal matrices:

$$S_1 = |3| \quad S_2 = \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} \quad S_3 = \begin{vmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{vmatrix}$$

Make a fibonacci guess for S_4 and verify you are right.

b) For this problem we use the formula for computing determinants of 2×2 and 3×3 matrices, and formula for recursion by cofactors to make a fibonacci guess for S_4 .

$$\det A_{2 \times 2} = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

$$\det A_{3 \times 3} = \begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a(ei - fh) - b(di - fg) + c(dh - eg)$$

recursion by cofactors: $D_n = 3D_{n-1} - D_{n-2}$

2 I came across this problem in the textbook and was intrigued by it since it related to topics we had covered thoroughly (determinants) and some we had only briefly glossed over (fibonacci sequence). When I initially tried to solve it, I mistakenly assumed we would have to create a matrix for S_4 , overcomplicating things, and also simply didn't understand how to incorporate fibonacci into this.

3 I began reading up on "Expansion of $\det A$ in Cofactors" from 4.3 of the textbook to understand more, which I include many steps of below. I also attempted similar but simpler practice problems from the section as well.

3 and 4

Let's say $\Rightarrow D_n = \det S_n$ (1)

Determinant S by cofactors $\Rightarrow D_n = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} + \dots = a_{11}C_{11} + a_{12}C_{12}$ (2)

Cofactor C_{ij} is the determinant of M_{ij} with the correct sign $\Rightarrow C_{ij} = (-1)^{i+j} M_{ij}$ (3); $M_{ij} = \det S_{n-1}$ (4)

from the provided $S_1, S_2, S_3 \dots$

$$a_{11} = 3$$

$$a_{12} = 1$$

$$a_{13}, a_{ij} = 0$$

from formula for $C_{ij} \dots$

$$C_{11} = (-1)^{1+1} \det S_{n-1}$$

$$= 1 \cdot \det S_{n-1}$$

$$= D_{n-1} \quad \text{from (1)}$$

$$C_{12} = (-1)^{1+2} \det S_{n-2}$$

$$= (-1) \det S_{n-2}$$

$$= -D_{n-2} \quad \text{from (1)}$$

put it all together $\Rightarrow D_n = a_{11}C_{11} + a_{12}C_{12} = D_n = 3D_{n-1} - D_{n-2}$

$$\det(S_1) = |3| = 3 \quad \det(S_2) = \begin{vmatrix} 3 & 1 \\ 1 & 3 \end{vmatrix} = (9) - (1) = 8 \quad \det(S_3) = \begin{vmatrix} 3 & 1 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 3 \end{vmatrix} = 3(9-1) - 1(3-0) + 0(1-0) = 3(8) - (3) = 21$$

From S_1, S_2, S_3 we have a pattern going of 3, 8, 21. To make a fibonacci guess for S_4 , we should look at the fibonacci sequence:

notice, our numbers are here!

$$F_n = 0, 1, 1, 2, \underline{3}, 5, \underline{8}, 13, \underline{21}, 34, 55, \dots; n=0, 1, 2, \dots$$

F_4
 $=$
 S_1

F_6
 $=$
 S_2

F_8
 $=$
 S_3

because $S_n; S_1, S_2, S_3, S_4$

Since this pattern is going by multiples of 2, lets figure out how to start with $n=1$ and get the correct pattern

Messing around a bit, we find $F_{2n+2} = S_n, n=1, 2, 3, 4$. Using this pattern... $S_4 = F_{10} = 55$.
Fibonacci guess for S_4 $\hat{J} = 55$

Verifying the correct answer, we use our $D_n = 3D_{n-1} - D_{n-2}$ for from before.

det S_4 :

$$\begin{aligned} D_4 &= 3D_3 - D_2 \\ &= 3(21) - (8) \\ &= 63 - 8 \\ &= 55 \end{aligned}$$

Looks like our Fibonacci guess was correct.

1
a) From 5.4 in the textbook, # 11:

From their trace and determinant, at what time t do the following matrices change between stable with real eigenvalues, stable with complex eigenvalues, and unstable?

b) For this problem we apply what we know about stability regarding real and complex eigenvalues with determinants and trace tests.

$$A_1 = \begin{bmatrix} 1 & -1 \\ -1 & -1 \end{bmatrix}, \quad A_2 = \begin{bmatrix} 0 & 4-t \\ 1 & 2 \end{bmatrix}, \quad A_3 = \begin{bmatrix} t & -1 \\ 1 & t \end{bmatrix}$$

2 Upon first attempting this problem, I fully understood how $\text{Re}(\lambda)$ affected the stability of a matrix, and how to properly apply the trace when necessary, but I was totally unsure of how to make sense of the complex eigenvalues in the same regard. I could not see the broader picture of the role complex eigenvalues play in graphing the stable and unstable regions for a 2×2 matrix.

3 From reading the section "Stability of differential equations" in 5.4, and reviewing my lecture notes where we briefly discussed the complex eigenvalue role in stability, I was able to deduce the following key points/notes pertinent for this problem:

Stability tests:

- $\text{Re}(\lambda_1) < 0 \Rightarrow$ The trace $a+d$ must be negative.
- $\text{Re}(\lambda_2) > 0 \Rightarrow$ The determinant $ad-bc$ must be positive

From 5.4: $\det \begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix} = \lambda^2 - (\text{trace})\lambda + \det = 0$

$\downarrow$
 Quadratic formula
 $\downarrow$
 $(\text{trace})^2 - 4\det$

$$\lambda = \frac{1}{2} [\text{trace} \pm \sqrt{(\text{trace})^2 - 4\det}]$$

$$(\text{trace})^2 - 4\det \geq 0 \quad (1)$$

Only for a 2×2 matrix:

- $\det(A) < 0 \Rightarrow$ Saddle point
- $\det(A) > 0$

$$\hookrightarrow \tau > 0 \Rightarrow \text{unstable}$$

$$\hookrightarrow \tau < 0 \Rightarrow \text{stable}$$

* Trace (τ) is the sum of the diagonal elements

4

$$A_1 = \begin{bmatrix} 1 & -1 \\ t & -1 \end{bmatrix}$$

Trace: $\tau = (1) + (-1) = 0$ neither purely stable or unstable

$$\begin{aligned} \det(A_1) &= (-1)(1) - (-1)(t) \\ &= -1 + t \\ &= t - 1 \end{aligned}$$

$$(\tau)^2 - 4 \det(A_1) \geq 0 \quad (1)$$

$$\begin{aligned} (0)^2 - 4(t-1) &\geq 0 \\ -4t + 4 &\geq 0 \\ 4 &\geq 4t \\ t &\leq 1 \end{aligned}$$

When $t=1$, $(1) = 0$

Stable with...

- ↳ Real λ : $t < 1$
- ↳ Complex λ : $t > 1$

A_1 changes stability

at $t=1$

$$A_2 = \begin{bmatrix} 0 & 4-t \\ 1 & -2 \end{bmatrix}$$

$\tau = (0) + (-2) = -2$: stable

$$\begin{aligned} \det(A_2) &= (0)(-2) - (4-t)(1) \\ &= 0 - (4-t) \\ &= t - 4 \end{aligned}$$

stable when $t > 4$

$$(-2)^2 - 4(t-4) \geq 0$$

$$\begin{aligned} 4 - 4t + 16 &\geq 0 \\ 20 &\geq 4t \\ t &\leq 5 \end{aligned}$$

When $t=5$, $(1) = 0$

Stable with...

- ↳ $\text{Re}(\lambda)$: $t < 5$
- ↳ $|\text{Im}(\lambda)|$: $t > 5$

A_2 changes stability

at $t=4$ and $t=5$

$$A_3 = \begin{bmatrix} t & -1 \\ 1 & t \end{bmatrix}$$

$\tau = (t) + (t) = 2t$: Stable: $t < 0$
unstable: $t > 0$

$$\begin{aligned} \det(A_3) &= (t)(t) - (-1)(1) \\ &= t^2 + 1 > 0 \quad \forall \mathbb{R} t \end{aligned}$$

$$(2t)^2 - 4(t^2 + 1) \geq 0$$

$$4t^2 - 4t^2 - 4 \geq 0$$

$$-4 \geq 0$$

never stable
with $\text{Re}(\lambda)$, only
complex

Stable with...

- ↳ $\text{Re}(\lambda)$: Never!
- ↳ $|\text{Im}(\lambda)|$: $t < 0$

A_3 changes stability

at $t=0$

1
a)

Singular Value Decomposition :

DEFN. Any $A_{m \times n}$ can be factored into

$$A = U \Sigma U^T = (\text{orthogonal}) (\text{diagonal}) (\text{orthogonal})$$

NOTES : Somewhat of a combination between LU gaussian elimination and QR gram-Schmidt orthogonalization.

Related to eigenvalue-vector factorization of a positive definite matrix.

Σ : diagonal rectangular matrix

↳ eigenvalues from $A^T A$ (sigma, σ) : Singular values of A

With rectangular matrices, we must take into account $A^T A$ and $A A^T$

U and V^T : any two orthogonal matrices (not necessarily transposes of each other)

b) How can we apply what we've learned to SVD application?

U and V provide us with orthonormal bases for the 4 fundamental spaces:

- first 'r' columns of U : $C(A)$
- last 'm-r' columns of U : $N(A^T)$
- first 'r' columns of V : $C(A^T)$
- last 'm-r' columns of V : $N(A)$

Other relations of importance:

- numerically stable computations
- vector length, least-squares, projection

SVD can help extend what we've learned about polar factorization to matrices of any size:

every real square matrix can be factored into $A = QS$, where Q is orthogonal, S is symmetric positive semidefinite.

proof

$$A = U \Sigma V^T = \underbrace{(U V^T)}_{V^T V = I} (V \Sigma V^T)$$

$$Q = UV^T$$

$$Q^T Q = (UV^T)^T (UV^T)$$

$$= V U^T U V^T = I$$

$$S = V \Sigma V^T$$

↓ symmetric and positive semidefinite

Since Σ is.

c) Find the SVD from the eigenvectors v_1, v_2 of $A^T A$ and $Av_i = \sigma_i u_i$.

(6.3) $A^T A = V \Sigma^T \Sigma V^T$; V must be the eigenvector matrix for $A^T A$. The eigenvalue matrix is $\Sigma^T \Sigma$; which is $n \times n$ with $\sigma_1^2, \dots, \sigma_r^2$ on the diagonal.

Fibonacci matrix: $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

Step 1 find $A^T A$

$$A^T A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1(1)+1(1) & 1(1)+1(0) \\ 1(1)+0(1) & 1(1)+0(0) \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

Step 2 find eigenvalues

$$\det(A^T A - \lambda I) = 0 \\ \det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 1-\lambda \end{pmatrix} = 0 \\ (2-\lambda)(1-\lambda) - 1(1)(1) = 0 \\ 2 - 2\lambda - \lambda + \lambda^2 - 1 = 0 \\ \lambda^2 - 3\lambda + 1 = 0 \\ \lambda = \frac{3 \pm \sqrt{5}}{2}$$

Step 3 Find σ_i from $\Sigma^T \Sigma$ and λ

$$\Sigma^T \Sigma = \begin{bmatrix} \sigma_1^2 & & \\ & \sigma_2^2 & \\ & & \ddots \end{bmatrix} \quad \lambda_i = \sigma_i^2 \\ \sqrt{\sigma_i^2} = \sigma_i = \sqrt{\lambda_i}$$

Step 5 compute u_i with $u_i = \frac{Av_i}{\sigma_i}$

$$u_1 = \frac{1}{\sqrt{\frac{3+\sqrt{5}}{2}}} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -\frac{1}{2}(1+\sqrt{5}) \end{bmatrix} \\ = \frac{\sqrt{5}-1}{2} \begin{bmatrix} 2 - \frac{1}{2}(1+\sqrt{5}) \\ 1 - \frac{1}{2}(1+\sqrt{5}) \end{bmatrix} \\ = \begin{bmatrix} (\sqrt{5}-1) - \frac{1}{4}(\sqrt{5}-1)(1+\sqrt{5}) \\ (\frac{\sqrt{5}-1}{2}) - \frac{1}{4}(\sqrt{5}-1)(1+\sqrt{5}) \end{bmatrix} \\ = \begin{bmatrix} \sqrt{5}-1-1 \\ \frac{\sqrt{5}-1}{2}-1 \end{bmatrix} \\ = \begin{bmatrix} \sqrt{5}-2 \\ \frac{\sqrt{5}-3}{2} \end{bmatrix}$$

Step 4 Compute eigenvectors from $(A^T A - \lambda I)\vec{v} = 0$

$$\begin{bmatrix} 2 - \frac{3+\sqrt{5}}{2} & 1 \\ 1 & 1 - \frac{3+\sqrt{5}}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \frac{1+\sqrt{5}}{2} & 1 \\ 1 & -\frac{1+\sqrt{5}}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \left(\frac{1+\sqrt{5}}{2} \right) v_1 + v_2 = 0 \quad \vec{v}_1 = \begin{bmatrix} 1 \\ -\frac{1}{2}(1+\sqrt{5}) \end{bmatrix} \\ \left(\frac{1+\sqrt{5}}{2} \right) v_1 = -v_2$$

$$\begin{bmatrix} 2 - \frac{3-\sqrt{5}}{2} & 1 \\ 1 & 1 - \frac{3-\sqrt{5}}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} \frac{1-\sqrt{5}}{2} & 1 \\ 1 & -\frac{1-\sqrt{5}}{2} \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ \left(\frac{1-\sqrt{5}}{2} \right) v_1 + v_2 = 0 \quad \vec{v}_2 = \begin{bmatrix} 1 \\ -\frac{1}{2}(1-\sqrt{5}) \end{bmatrix} \\ \left(\frac{1-\sqrt{5}}{2} \right) v_1 = -v_2$$

Step 6 Compute $A = U \Sigma V^T$; $U = [u_1 \ u_2]$

$$A = \underbrace{\begin{bmatrix} \sqrt{5}-2 & \sqrt{5}+2 \\ \frac{\sqrt{5}-3}{2} & \frac{\sqrt{5}+3}{2} \end{bmatrix}}_U \underbrace{\begin{bmatrix} \sqrt{\frac{3+\sqrt{5}}{2}} & 0 \\ 0 & \frac{3-\sqrt{5}}{2} \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} 1 & -\frac{1}{2}(1+\sqrt{5}) \\ 1 & -\frac{1}{2}(1-\sqrt{5}) \end{bmatrix}}_{V^T}$$

$$u_2 = \frac{1}{\sqrt{\frac{3-\sqrt{5}}{2}}} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -\frac{1}{2}(1-\sqrt{5}) \end{bmatrix} \\ = \frac{\sqrt{5}+1}{2} \begin{bmatrix} 2 - \frac{1}{2}(1-\sqrt{5}) \\ 1 - \frac{1}{2}(1-\sqrt{5}) \end{bmatrix} \\ = \begin{bmatrix} (\sqrt{5}+1) - \frac{1}{4}(\sqrt{5}+1)(1-\sqrt{5}) \\ \frac{1}{2}(\sqrt{5}+1) - \frac{1}{4}(\sqrt{5}+1)(1-\sqrt{5}) \end{bmatrix} \\ = \begin{bmatrix} \sqrt{5}+1+1 \\ \frac{1}{2}(\sqrt{5}+1)+1 \end{bmatrix} \\ = \begin{bmatrix} \sqrt{5}+2 \\ \frac{\sqrt{5}+3}{2} \end{bmatrix}$$

REFERENCES

- Lecture Notes
- Discussion Worksheets
- Linear Algebra and its Applications (textbook)
- bmcbgenomics.biomedcentral.com
- biochem218.stanford.edu